



National University
of computer and emerging sciences

Deep Learning for Perception

COURSE INSTRUCTOR: MUHAMMAD SAIF UL ISLAM

Lecture Outline – Week#1

- Introductory words
- Introduction to the Course
- Discussion on Course outline
- Course plan, Assignments and Project
- Machine Learning Recap
- Introduction to Deep Learning and Neural networks

About Myself

Muhammad Saif ul Islam

Education:

PhD Scholar (Computer Science)

- FAST-NUCES, LHR



Masters in Data Science - 2019

- FAST-NUCES, KHI



Bachelors in Computer Science -2017

- Bahria University, KHI



Work Experience: 9+ Years

IT Instructor – 5 Months

- IBA-BBSYDP



Python Developer – 7 Months

- Innovative Solutions



Sr. Operations Engineer – 1 Year

- Gfk Etilize



Lecturer – 2.5 years

- Mohammad Ali Jinnah University



Lecturer – 6 Months

- Beaconhouse National University



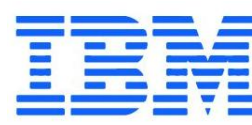
Lecturer – Since Spring 2023

- FAST-NUCES



About Myself

Certifications



25+

Sr. No.	Certification Year	Title	Institute	Sr. No.	Certification Year	Title	Institute
1	August 2026	Introduction to Generative AI: Concepts and Techniques	Alberta Machine Intelligence Institute	14	August 2023	Programming for Everybody (Getting Started with Python)	University of Michigan
2	July 2026	Foundations: Data, Data, Everywhere	Google	15	August 2023	Python Data Structures	University of Michigan
3	September 2023	Exploratory Data Analysis With Python and Pandas	Coursera	16	August 2023	Python Project for Data Science	IBM
4	August 2023	Python for Everybody Specialization	University of Michigan	17	August 2023	Applied Plotting, Charting & Data Representation in Python	University of Michigan
5	August 2023	Python for Data Analysis: Pandas & NumPy	Coursera	18	August 2023	Image Processing with Python	Coursera Projects
6	August 2023	Python Project: Software Engineering and Image Manipulation	University of Michigan	19	October 2018	Querying Data with Transact-SQL	Microsoft
7	August 2023	Image Segmentation with Python and Unsupervised Learning	Coursera Projects	20	June 2018	Applied Machine Learning in Python	University of Michigan
8	August 2023	Python Classes and Inheritance	University of Michigan	21	September 2017	Introduction to Data Science in Python	University of Michigan
9	August 2023	Python Functions, Files, and Dictionaries	University of Michigan	22	April 2017	HTML5 Introduction	W3Cx
10	August 2023	Python Basics	University of Michigan	23	April 2017	Data Science Essentials	Microsoft
11	August 2023	Capstone: Retrieving, Processing, and Visualizing Data with Python	University of Michigan	24	April 2017	Introduction to Python for Data Science	Microsoft
12	August 2023	Using Databases with Python	University of Michigan	25	March 2017	Introduction to R for Data Science	Microsoft
13	August 2023	Using Python to Access Web Data	University of Michigan	26	March 2017	Programming with Python for Data Science	Microsoft

About Myself

Publications

- 1-Rizwan, M., Wali, A., Islam, M. S. U., & Fatima, S. (2026). Fast and Accurate Class-Level Machine Unlearning using Impair-Repair and Noise-Induced Forgetting Mechanisms. Cognitive Computation. DOI: 10.1007/s12559-026-10597-4
- 2-Islam, M. S. U., & Wali, A. (2026). Federated transfer learning for generalized salt body segmentation across multiple heterogeneous seismic datasets. Discover Geoscience. DOI: 10.1007/s44288-026-00656-6
- 3-Ahmad, I. M., Ashar, B., Munir, S., Wali, A., & Islam, M. S. U. (2026). A hybrid deep learning framework for epileptic seizure prediction using scalp and intracranial EEG data. Computerized Medical Imaging and Graphics. DOI: 10.1016/j.compmedimag.2026.102759
- 4-Islam, M. S. U., & Wali, A. (2026). GeoFedNet: Federated learning for privacy-aware, robust, and generalizable seismic interpretation. Computers & Geosciences.
- 5-Islam, M. S. U., & Wali, A. (2025). Weighted ensemble transfer learning with EfficientNet: Advancing salt body segmentation in seismic imaging. Computational Geosciences.
- 6-Islam, M. S. U. (2025). Robust Flood Area Segmentation through Federated Learning on Distributed Remote Sensing Data. 2025 Horizons of Information Technology and Engineering (HITE).

About Myself

Publications

7-Islam, M. S. U., & Wali, A. (2025). Enhancing generalization in seismic image segmentation through multi-dataset integration and transfer learning. Journal of Earth System Science.

8-Islam, M. S. U. (2025). Comparative Analysis of Data Augmentation Techniques for Enhancing Generalizability in Brain Tumor Classification with Limited Datasets. 2025 International Conference on Emerging Technologies in Electronics, Computing, and Communication (ICETECC).

9-Islam, M. S. U., & Wali, A. (2024). A comprehensive review of deep learning techniques for salt dome segmentation in seismic images. Journal of Applied Geophysics.

10-Khan, M. M., Islam, M. S. U., Siddiqui, A. A., & Qadri, M. T. (2023). Dual deterministic model based on deep neural network for the classification of pneumonia. Intelligent Decision Technologies. DOI: 10.3233/idt-220192

11-Mehboob, M., Ali, M. S., Islam, S. U., & Ali, S. S. (2022). Evaluating Automatic CV Shortlisting Tool For Job Recruitment Based On Machine Learning Techniques. 2022 Mohammad Ali Jinnah University International Conference on Computing (MAJICC).

12-Islam, M. S. U. (2020). Using deep learning based methods to classify salt bodies in seismic images. Journal of Applied Geophysics. DOI: 10.1016/j.jappgeo.2020.104054

13-Islam, M. S. U. (2017). Rating visual contents of website using brain computer interface. 2017 International Conference on Information and Communication Technologies (ICICT).

Students' Introduction



Name?

Expectation:

- Why did you choose this course?
- What do you want to learn in this course?

Course Outline



Course plan, Assignments and Quizzes

	Graded Assessment types	Weights (%)
1	Project + Research paper	15%
2	Quiz (3), No retake	5%
3	Assignments + Class Activity, No retake	10%
4	Mid Exam (2)	30%
5	Final Assessment	40%
	Total:	100%

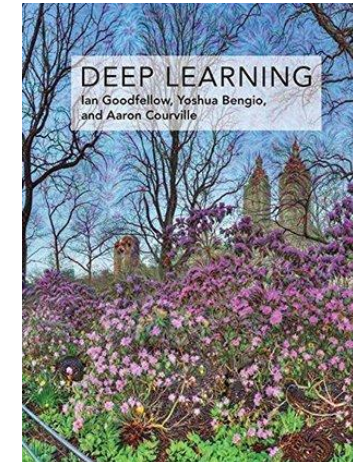
Late Submission Policy

1. **Quiz:** You lose 1 mark for every minute you submit your quiz past the deadline. For example, if you submit your quiz 5 minutes late, you lose 5 marks from your final score.
2. **Assignment:** You lose 15% of the maximum marks for every day you submit your assignment late.
3. **Project:** You lose 10% of the maximum marks for every day you submit your assignment late.
4. **Retake Policy:** There are no retakes for any assessments.

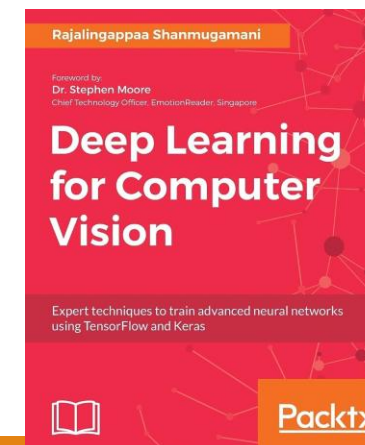
Resources

Books:

Deep Learning by Yoshua Bengio



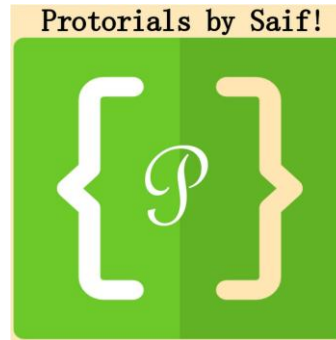
Deep Learning for Computer Vision



Resources

Protorials By Saif

@protorialsbysaif



Consultation

Office Hours:

Office#58, F Block

Tue/Thurs - 9:00AM-11:00AM

Email: saif.islam@lhr.nu.edu.pk



Machine Learning Recap

- **The Problem:** You need to sort harvested apples into two boxes: "Premium" and "Standard".

Apple 1: Red, 160 grams = Premium

Apple 2: Green, 120 grams = Standard

Apple 3: Red, 155 grams = Premium

Apple 4: Green, 170 grams = Standard

Apple 5: Red, 110 grams = Standard

Apple 6: Red, 180 grams = Premium



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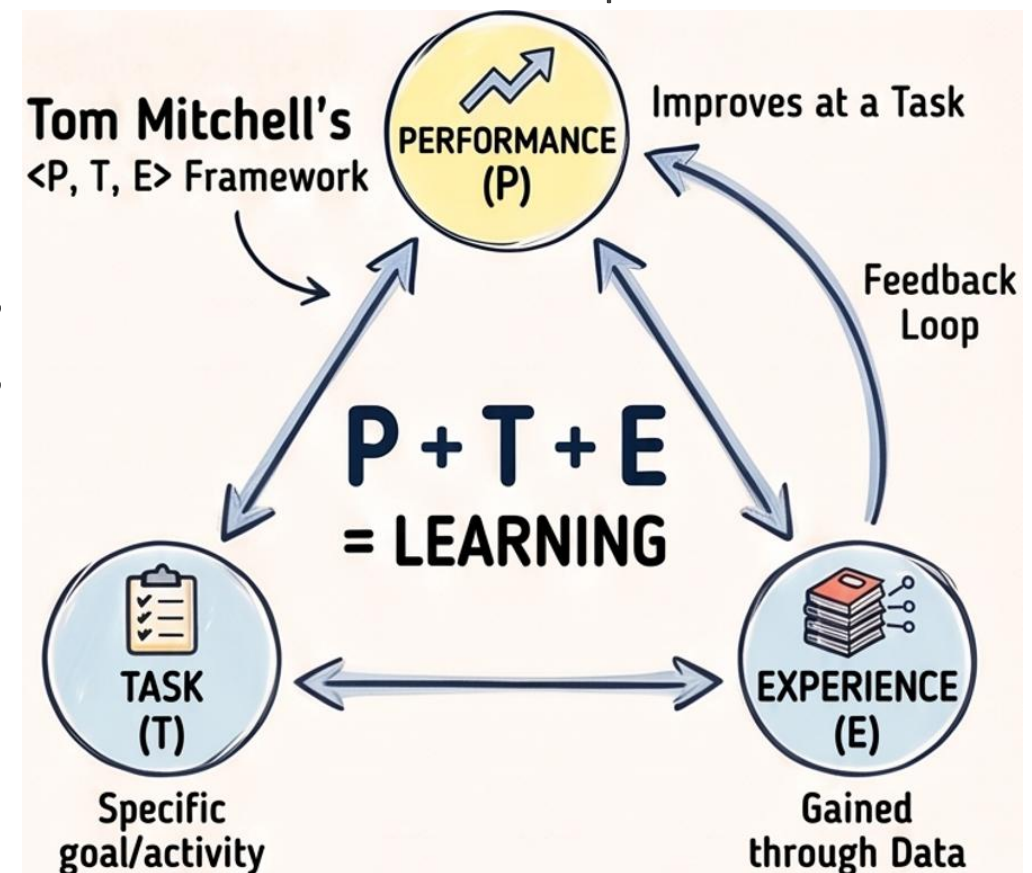
Predict?
Red, 152 grams

The Machine Learning Approach

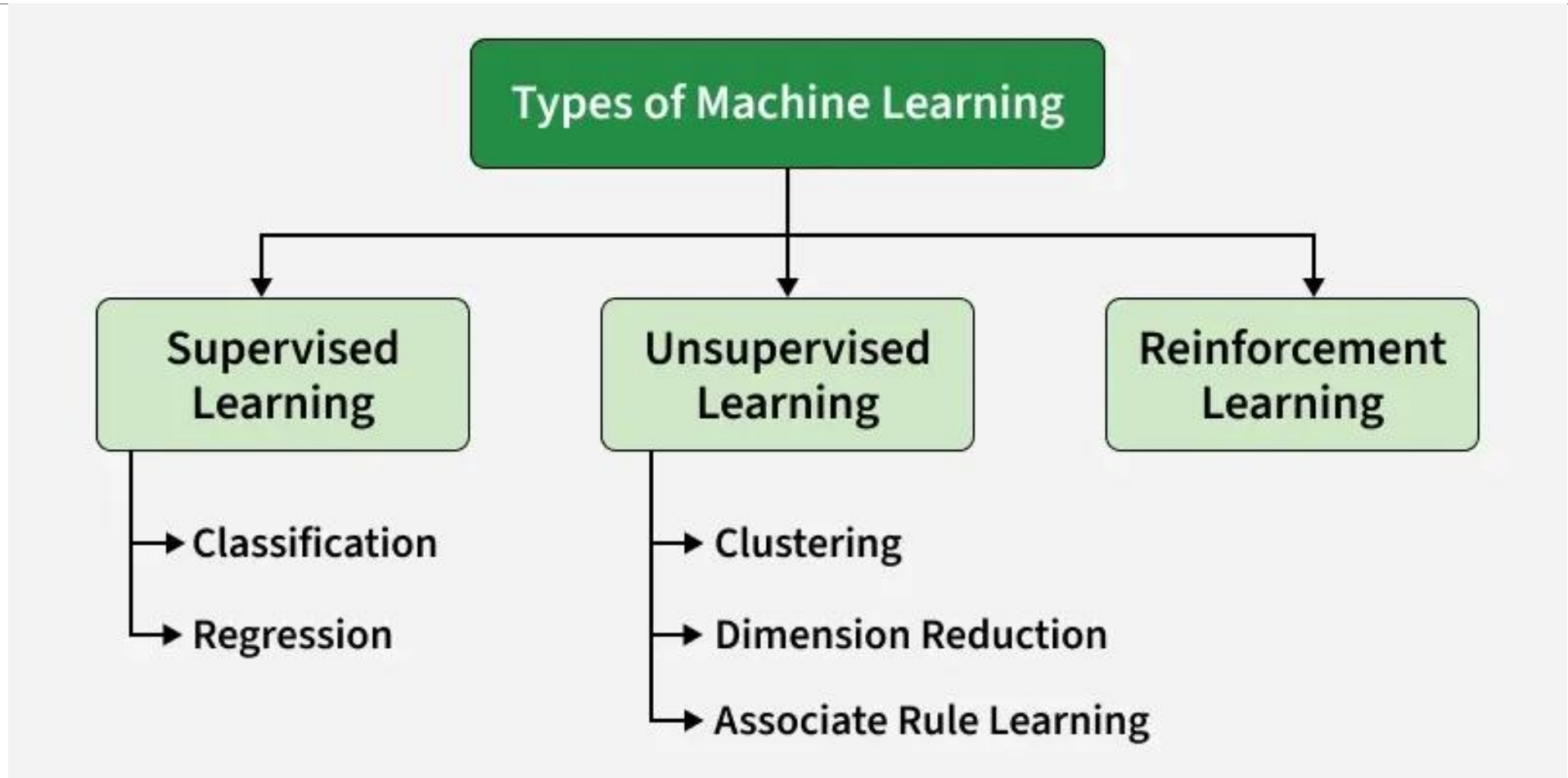
In machine learning, you do not write the rules. You provide historical data. The computer learns the rules by finding patterns in your examples.

“A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E .”

Tom M. Mitchell (1997)



Types of Machine Learning



Types of Machine Learning

Supervised Learning



Apple



Orange



Mango

Labeled Data

Unsupervised Learning



Unlabeled Data

Types of Machine Learning

Supervised Learning

- Model is trained on **labeled data**.
- **Regression**: Predicting a numerical value
- **Classification**: Predicting a categorical label

Unsupervised Learning

- Model is trained on **unlabeled data**.
- **Clustering**: Grouping similar data points together.
- **Dimensionality Reduction**: Reducing the number of features in a dataset

Machine Learning Process

